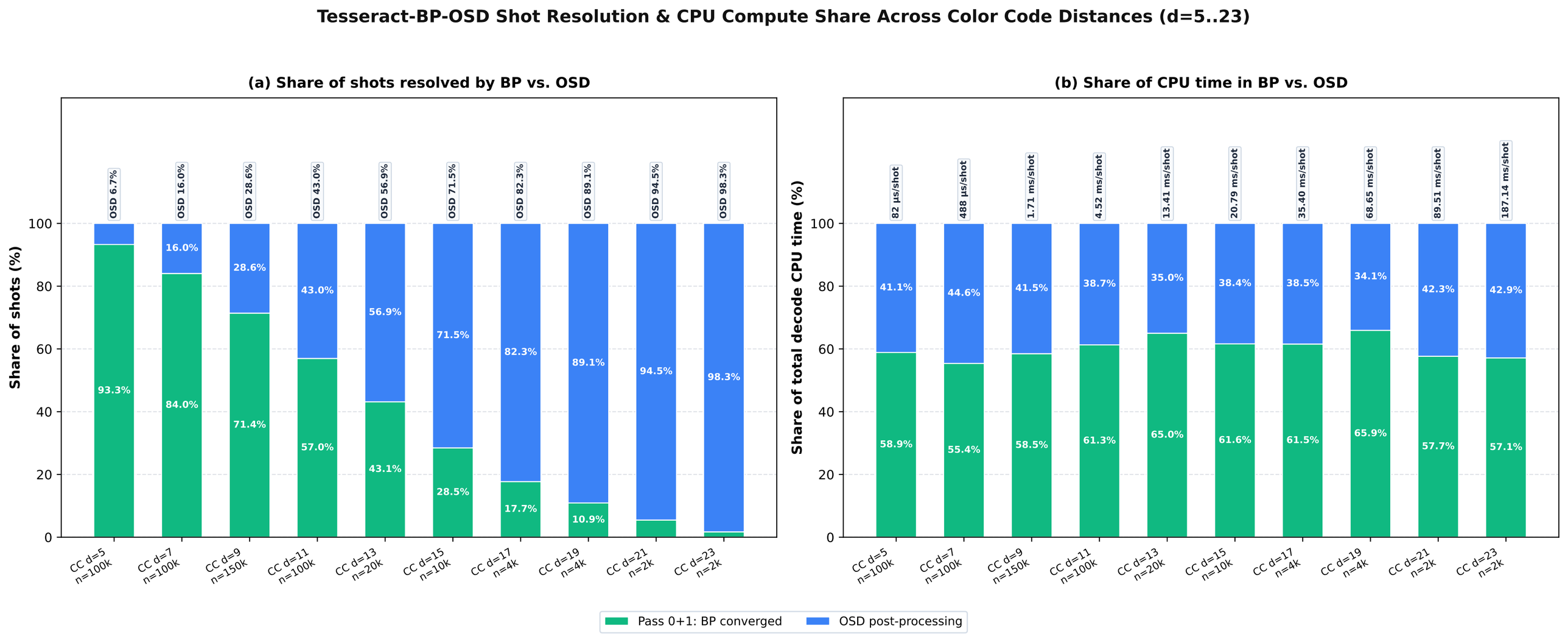
Tesseract-BP-OSD Shot Resolution & CPU Compute Share Across Color Code Distances (d=5..23)
(a) Share of shots resolved by BP vs. OSD
Share of shots (%)
OSD 6.7%
OSD 16.0%
OSD 28.6%
OSD 43.0%
OSD 56.9%
OSD 71.5%
OSD 82.3%
OSD 89.1%
OSD 94.5%
OSD 98.3%
93.3%
84.0%
71.4%
57.0%
43.1%
28.5%
17.7%
10.9%
94.5%
98.3%
CC d=5 n=100k
CC d=7 n=100k
CC d=9 n=150k
CC d=11 n=100k
CC d=13 n=20k
CC d=15 n=10k
CC d=17 n=4k
CC d=19 n=4k
CC d=21 n=2k
CC d=23 n=2k
(b) Share of CPU time in BP vs. OSD
Share of total decode CPU time (%)
82 µs/shot
488 µs/shot
1.71 ms/shot
4.52 ms/shot
13.41 ms/shot
20.79 ms/shot
35.40 ms/shot
68.65 ms/shot
89.51 ms/shot
187.14 ms/shot
41.1%
44.6%
41.5%
38.7%
35.0%
38.4%
38.5%
34.1%
42.3%
42.9%
58.9%
55.4%
58.5%
61.3%
65.0%
61.6%
61.5%
65.9%
57.7%
57.1%
Pass 0+1: BP converged
OSD post-processing